



PYTHON BOOTCAMP

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DICTIONARIES



Dictionaries:

- Key-value data structure
- Key need to be unique
- Value can be duplicated

```
3 student = {
4     "name": "Alice",
5     "age": 20,
6     "grade": "A",
7     "courses": ["Math", "Science", "English"]
8 }
9
10 # Accessing and modifying
11 print(student["name"])           # "Alice"
12 print(student.get("age"))        # 20
13 student["age"] = 21              # Modify value
14 student["email"] = "alice@email.com" # Add new key-value
```

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Dictionaries Method:

```
17 keys = student.keys()      # Get all keys
18 values = student.values()  # Get all values
19 items = student.items()    # Get key-value pairs
20
21 print(keys)
22 print(values)
23 print(items)
```

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Iterating Dictionaries:

```
25 # Iterating through dictionaries
26 for key in student:
27     print(f"{key}: {student[key]}")
28
29 for key, value in student.items():
30     print(f"{key}: {value}")
```

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Nested Dictionaries:

```
32 # Nested dictionaries
33 company = {
34     "employees": {
35         "john": {"age": 30, "department": "IT"},
36         "jane": {"age": 25, "department": "HR"}
37     },
38     "departments": ["IT", "HR", "Finance"]
39 }
40
41 print(company["employees"].items())
42 print(company["departments"])
```

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Exercises:

1. Create a dictionary called `student_records` with the following information:
"student_001": name is "John", age is 19, major is "Computer Science", grades are [85, 92, 78]
"student_002": name is "Sarah", age is 20, major is "Biology", grades are [90, 88, 95]
2. Add a new student "student_003" with name "Mike", age 18, major "Math", grades [82, 79, 91]
3. Update John's age to 20
4. Loop through the dictionary and print each student's information in this format:
"Student ID: [id], Name: [name], Major: [major]"